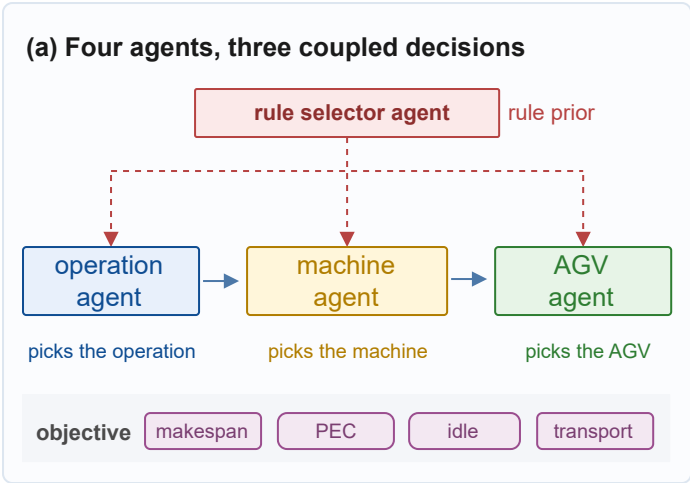
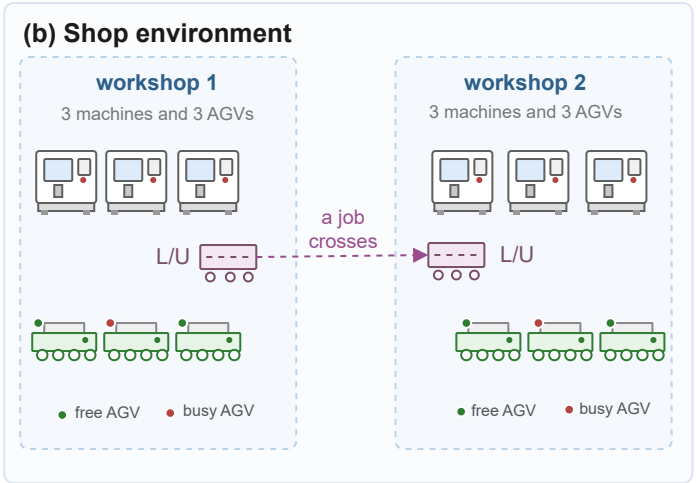
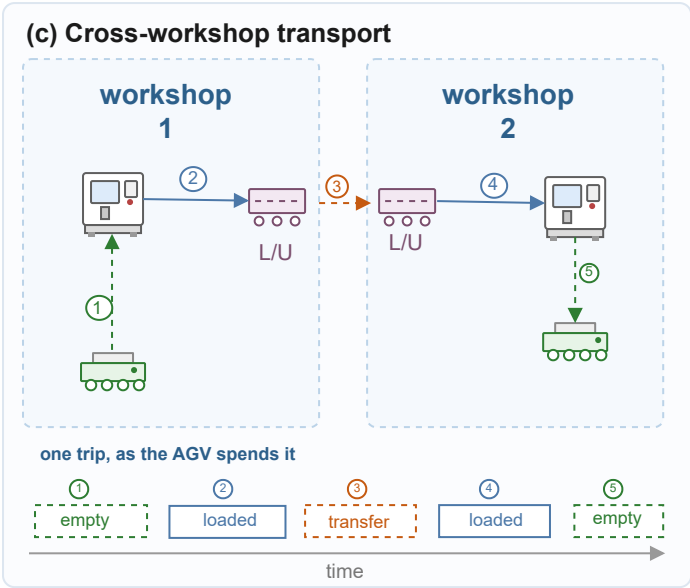
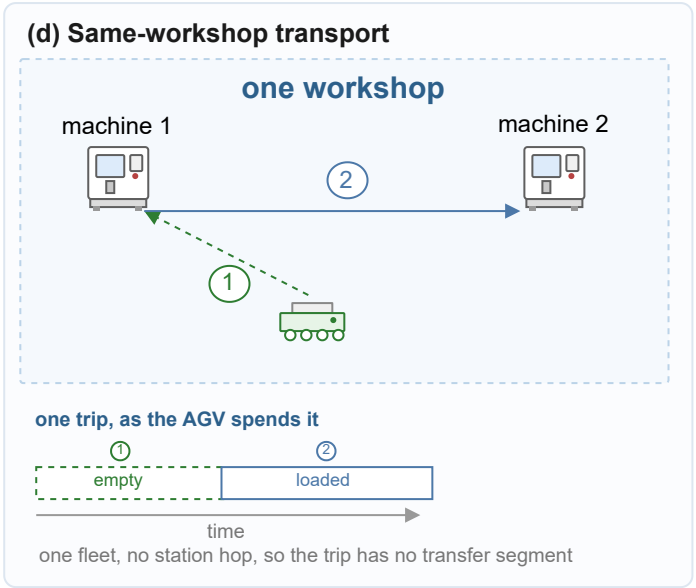

(a) Four agents, three coupled decisions
rule selector agent
rule prior
operation agent
machine agent
AGV agent
picks the operation
picks the machine
picks the AGV
objective
makespan
PEC
idle
transport


(b) Shop environment
workshop 1
3 machines and 3 AGVs
workshop 2
3 machines and 3 AGVs
a job crosses
L/U
L/U
free AGV
busy AGV
free AGV
busy AGV


(c) Cross-workshop transport
workshop 1
workshop 2
L/U
L/U
one trip, as the AGV spends it
empty
loaded
transfer
loaded
empty
time


(d) Same-workshop transport
one workshop
machine 1
machine 2
one trip, as the AGV spends it
empty
loaded
time
one fleet, no station hop, so the trip has no transfer segment